Step 1: Python kya hai

Python ek Scripting Language hai. Editor: PyCharm

Interpreted: code ko line by line chalata hai

Interactive: ek command likho, turant output milta hai

Object-oriented: program ko objects ke form me organize kar sakte hain

Class = kisi object ka blueprint/design

Object = class se banaya gaya actual item

Script Mode: program .py file me save karke poori file run karte hain

Bytecode: hello.py → Python bytecode me convert karta hai → hello.pyc

Step 2: print, comment, variable (NEW)

python

```
print("Hello") # screen par output dikhata hai

# ye comment hai, Python isko ignore karta hai

name = "Rinku" # name = variable, "Rinku" = value
age = 20
```

Variable = data store karne ka dabba

Variable ka naam number se start nahi hota, aur space nahi hota (my_name sahi hai)

Step 3: Data Types (NEW)

Type Matlab

Example

String (str) Characters ka set (" ") "Raj"

Integer (int) Poora number 25

Float (float) Decimal number 5.5

Boolean (bool) True ya False True

type(25) likhne se type pata chalta hai.

Step 4: input aur Operators (NEW)

python

name = input("Naam likho: ")

age = int(input("Age likho: ")) # input hamesha string aata hai, isliye int() lagaya

Math operators: + - * / % (bacha hua hissa) ** (power) // (poora bhaag)

Compare operators: == (barabar) != (barabar nahi) > < >= <=

Step 5: If / Elif / Else

Condition ke True ya False hone ke hisaab se decision leta hai.

if = pehli condition

elif = dusri/teesri condition (multiple conditions check karne ke

liye)

else = koi bhi condition True nahi hai

python

marks = 75

if marks >= 80:

print("A grade")

elif marks >= 60:

print("B grade")

else:

print("Fail")

Dhyan rakho (NEW): : lagana zaroori hai, aur andar wali line me indentation (4 space) dena zaroori hai.

Step 6: While Loop

Jab tak condition True hai, loop chalega. Condition False hote hi ruk jayega.

python

i = 1

while i <= 5:

print(i)

i = i + 1 # ye bhool gaye to loop kabhi nahi

rukega

Output: 1 2 3 4 5

Step 7: For Loop

Sequence (list, range, string) ke elements ko ek-ek karke process karta hai.

List:

```
python
fruits = ["Apple", "Mango", "Banana"]
for fruit in fruits:
print(fruit)
```

Range:

```
python
for i in range(1, 6):
print(i)
```

Sudhar: range(1, 6) me 1 se 5 tak aata hai, 6 nahi aata. Last number hamesha chhoot jata hai. Aapne notes me 1 to 6 likha hai, wo galat

hai.

String:

python
name = "Rinku"
for letter in name:
print(letter)

Step 8: Break aur Continue

break = loop ko poori tarah band
continue = current iteration skip karo, aage chalo

python
for i in range(1, 6):
if i == 3:
break
print(i)
Output: 1 2

python
for i in range(1, 6):
if i == 3:
continue
print(i)
Output: 1 2 4

5

Step 9: List

Ek variable me multiple values store karne ke liye. Square brackets [ ] me likhte hain. Alag-alag type ka data ek saath rakh sakte hain.

python

```
data = ["Raj", 25, 5.5, True]
fruits = ["Apple", "Mango", "Banana"]
print(fruits[0]) # Apple (index 0 se shuru hota hai)
print(fruits[-1]) # Banana (-1 = last item) (NEW)
```

List ki value change ho sakti hai.

List Functions

#	Function	Kya karta hai	Example
1	append()	End me new item add	fruits.append("Grapes")
2	insert()	Specific index par item add	fruits.insert(1, "Orange")
3	remove()	Value ko remove	fruits.remove("Mango")
4	sort()	Ascending order me arrange	numbers.sort()
5	reverse()	Order ulta	numbers.reverse()
6	count()	Value kitni baar aayi	numbers.count(10)
7	index()	Value ka index number	

fruits.index("Mango")

8 max() Sabse bada number max(numbers)

9 min() Sabse chhota number min(numbers)

10 list() Sequence ko list me convert list("Rakesh")

python

```
numbers = [10, 20, 50, 80]
print(max(numbers)) # 80
x = list("Rakesh")
print(x) # ['R','a','k','e','s','h']
```

Extra (NEW): len(fruits) se items ki ginti pata chalti hai, pop() last item nikal deta hai.

Step 10: Tuple

Ek variable me multiple values store karna, lekin baad me values change nahi kar sakte. Round brackets ( ) me likhte hain.

python

```
fruits = ("Apple", "Mango", "Banana")
print(fruits[0]) # Apple
```

List vs Tuple: List [ ] badal sakti hai, Tuple ( ) nahi badal sakta.

Step 11: Dictionary

(NEW)

Data key : value ki jodi me store hota hai. Curly brackets { } me likhte hain.

python
student = {"name": "Rinku", "age": 20}
print(student["name"]) # Rinku
student["age"] = 21 # value change
student["city"] = "Indore" # new item add
Step 12: Functions (NEW)

Function = code ka ek block jo ek baar likho aur baar-baar use karo.

def se function banate hain
Function ko call karne par hi wo chalta hai

Simple function:

python
def hello():
print("Hello Friends")

hello() #

call

Parameter ke saath (bahar se value dena):

python
def greet(name):
print("Hello", name)

greet("Rinku") # Hello Rinku

Return ke saath (result wapas dena):

python
def add(a, b):
return a + b

result = add(10, 20)
print(result) # 30

Default value:

python
def greet(name="Dost"):
print("Hello",

name)

greet() # Hello Dost
greet("Raj") # Hello Raj
Parameter = function banate time wala variable (a, b)
Argument = call karte time di gayi value (10, 20)
print sirf dikhata hai, return value wapas deta hai
Step 13: Date and Time (NEW)

Iske liye datetime module use karte hain. Module = ready-made code ka dabba, jise import karke use karte hain.

python
import datetime

now = datetime.datetime.now()
print(now) # 2026-10-07 22:45:30.123456

Sirf aaj ki date:

python
today = datetime.date.today()
print(today) # 2026-10-07

Alag-alag part

nikalna:

```
python
print(now.year) # 2026
print(now.month) # 10
print(now.day) # 7
print(now.hour) # 22
print(now.minute) # 45
print(now.second) # 30
```

Format badalna (strftime):

```
python
print(now.strftime("%d-%m-%Y")) # 07-10-2026
print(now.strftime("%H:%M:%S")) # 22:45:30
print(now.strftime("%d %B %Y")) # 07 October 2026
```

Code Matlab

%d Din (07)

%m Mahina number (10)

%B Mahine ka naam (October)

%Y Saal (2026)

%H Ghanta (24 hour)

%M Minute

%S Second

%A Din ka naam